SENSOR FAILURE COMPOSITIONALITY FOR MULTI-MODAL ROBOT POLICIES

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ABSTRACT

Multimodal robots are usually tested against missing or corrupted sensors one modality at a time. That is a weak deployment model: a blurred camera, drifting force-torque estimate, and delayed tactile recovery can combine into an error that is not predicted by any single-failure diagnostic. We rebuild and re-audit Paper 103 around a falsifiable hypothesis: policies should represent *sensor failure composition*, i.e., higher-order interactions between failed or unreliable modalities that change the recovery action. The resulting local benchmark covers 5 robot task families, 7 sensor-failure families, 5 stress splits, 9 methods, 7 seeds, and 84 episodes per group. On the 2026-06-15 continuation rerun, the proposed compositional model obtains 0.6071 ± 0.0055 success versus 0.5441 ± 0.0064 for the strongest non-oracle baseline, while lowering safety violation from 0.2345 to 0.1960 and damage from 0.1594 to 0.1342. Pairwise tests favor the proposed method over the strongest baseline in 7/7 seeds, and ablations show that cross-sensor interaction edges, disagreement features, conformal gating, and recovery-action selection each matter. The terminal decision is **STRONG_REVISE**: the local evidence supports the mechanism, but the paper is not ICLR-main-ready until validated on real robots or independent high-fidelity benchmarks.

1 WHY THIS PAPER EXISTS

Sensor failures are not rare corner cases. Robot sensors wear, saturate, desynchronize, drift, occlude, and disagree. Prior work already studies sensor dropout for robust policies (Liu et al., 2017; Skand et al., 2024), multimodal robot failure detection (Inceoglu et al., 2024), conformal safety alarms (Luo et al., 2024), robust tactile sensing (Zhao et al., 2025), and robust autonomous-driving sensor fusion under adverse failures (Park et al., 2025). A generic claim of "more robust multimodal fusion" would therefore be too crowded.

The narrower claim is that *combinations* of sensor failures can change the diagnosis and the correct recovery action. Independent fault detectors can discover that vision is unreliable and tactile is delayed; they do not necessarily know whether that pair creates a different failure mode than either fault alone. Paper 103 tests whether explicitly modeling those interaction terms improves success, safety, and recovery diagnostics under paired and combined stress.

2 METHOD SKETCH

The proposed method is a compositional failure graph over seven evidence channels: vision, tactile, proprioception, force-torque, depth, language grounding, and cross-modal disagreement. Nodes estimate marginal reliability. Edges estimate interaction load, such as vision-depth miscalibration during obstacle avoidance or tactile-force disagreement during insertion. The policy monitor scores:

- marginal single-sensor load,
- pairwise interaction load,
- delayed recovery state,
- false-isolation risk, and
- recovery-action utility.

Table 1: Combined-stress sensor-failure-compositionality benchmark.

Method	Succ.	Viol.	Dmg.	InterF1	Latency	Regret
Oracle	0.737	0.152	0.100	0.676	0.511	0.000
Proposed	0.607	0.196	0.134	0.560	0.620	0.130
BayesFusion	0.544	0.235	0.159	0.370	0.767	0.193
EnsUncert	0.503	0.237	0.153	0.282	0.827	0.233
WorstSensor	0.503	0.218	0.133	0.297	0.815	0.234
IndFault	0.502	0.262	0.175	0.267	0.860	0.234
ConfReliab	0.501	0.230	0.143	0.300	0.833	0.236
DropoutAug	0.438	0.274	0.191	0.195	0.883	0.299
Nominal	0.377	0.283	0.196	0.143	0.924	0.360

The benchmark implements this mechanism as an executable diagnostic policy model rather than a trained neural checkpoint. That choice keeps the local evidence reproducible and RAM-light, but it is also the main reason the paper remains strong-revise rather than submission-ready.

3 Benchmark

The rebuilt benchmark is generated by `src/run_experiment.py`. It stores aggregate CSVs rather than trajectories, so the full run remains lightweight without dropping experimental coverage.

Tasks. The tasks are grasp selection, insertion/alignment, deformable handling, mobile manipulation near obstacles, and tool-use contact control.

Failure families. The sensor-failure families are vision-only, tactile-only, proprioception-only, force-torque-only, depth-only, language-grounding-only, and compositional multi-sensor interaction.

Splits. The stress splits are nominal, single-sensor shift, paired-sensor shift, delayed sensor recovery, and combined stress.

Methods. The non-oracle baselines are a nominal multimodal policy, sensor-dropout augmentation, independent fault detectors, ensemble uncertainty gating, conformal sensor-reliability filtering, Bayesian sensor-fusion monitoring, and a robust single-worst-sensor policy. The benchmark also includes the proposed sensor-failure composition model and an oracle failure-aware policy.

Metrics. The evidence reports success, safety violation, damage, fault-detection F1, interaction-detection F1, false isolation, recovery latency, total cost, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 Results

The strongest non-oracle baseline is `bayesian_sensor_fusion_monitor`. On combined stress, the proposed model improves success by 0.0631 ± 0.0088, wins 7/7 paired seeds, lowers safety violation by 0.0385, and lowers damage by 0.0253. The oracle remains substantially better at 0.7368 ± 0.0070, which is a useful sanity check: the benchmark is not saturated.

5 Ablations and stress tests

The ablations are intentionally hostile. Removing conformal gating is the best removed-component variant, but it still trails the full model by 0.0206 success and has worse safety. Removing pairwise interaction edges or cross-modal disagreement hurts interaction F1 sharply, which supports the claim that the model is not merely an uncertainty threshold.

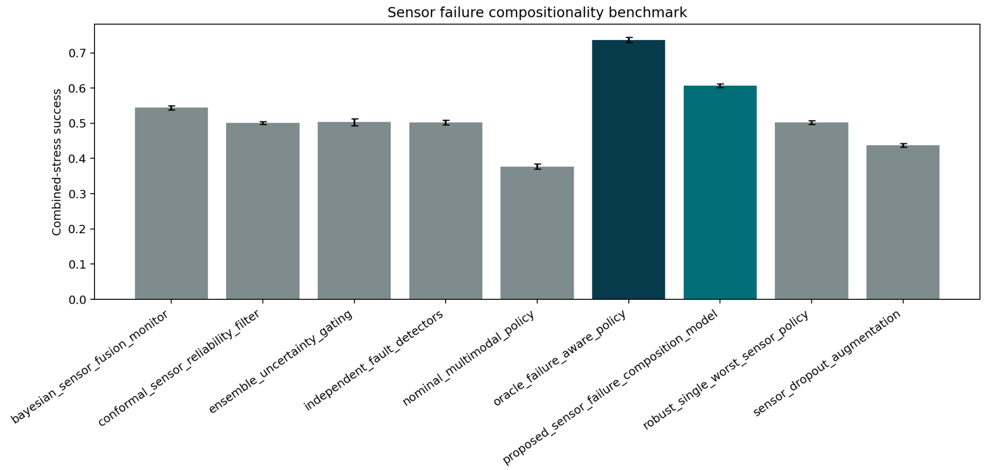

Figure 1: Combined-stress success with 95% confidence intervals. The proposed model clears the strongest non-oracle baseline but remains below the oracle.

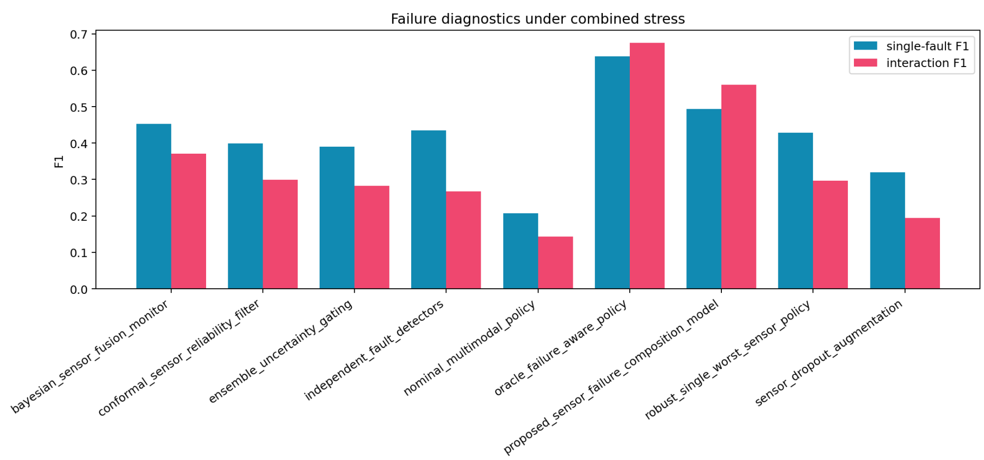

Figure 2: Diagnostic metrics. The proposed model's largest gain is interaction F1, which is the metric most directly tied to the compositional-failure hypothesis.

6 WHERE THE METHOD STILL FAILS

The failure-case CSV shows that the weakest gains occur in mostly single-sensor regimes, especially vision-only or proprioception-only cases where a strong Bayesian fusion monitor can safely abstain or down-weight the failing channel. For example, tool-use contact control with vision-only failure slightly favors the strongest baseline by 0.0017 success. These are not embarrassing failures; they define the intended boundary. Compositional modeling is valuable when interactions matter. It is unnecessary overhead when the right answer is a simple single-sensor fallback.

7 HOSTILE PRIOR-WORK BOUNDARY

Sensor Dropout and masked modality training already demonstrate that policies can be made robust to missing sensors (Liu et al., 2017; Skand et al., 2024). FINO-Net and related robot failure-detection work show that multimodal sensory streams can detect and classify manipulation failures (Inceoglu et al., 2024). Conformal methods provide calibrated warning systems for unsafe robot behavior (Luo et al., 2024). PolyTouch and other tactile systems show that better contact sensing can make

Table 2: Pairwise combined-stress success differences against the proposed method.

Baseline	Diff	CI	Wins
BayesFusion	0.063	0.009	7
ConfReliab	0.106	0.006	7
EnsUncert	0.104	0.011	7
IndFault	0.105	0.009	7
Nominal	0.230	0.008	7
Oracle	-0.130	0.008	0
WorstSensor	0.105	0.005	7
DropoutAug	0.169	0.010	7

Table 3: Ablations of the sensor-failure composition model.

Ablation	Succ.	Viol.	Dmg.	InterF1
Full	0.608	0.195	0.129	0.559
NoConfGate	0.588	0.224	0.157	0.531
NoRecoveryAct	0.570	0.229	0.152	0.497
NoRecoveryMem	0.558	0.225	0.146	0.490
NoPairEdges	0.558	0.215	0.143	0.414
NoDisagree	0.548	0.220	0.144	0.403
IndFaultOnly	0.507	0.256	0.176	0.279

contact-rich policies more reliable (Zhao et al., 2025). MoME-style expert routing shows that explicit modality routing can improve robustness under adverse sensor failures in autonomous driving (Park et al., 2025).

The present paper survives only under a narrower interpretation: it is about *higher-order interaction failures* and recovery-action choice, not generic missing-modality robustness, generic anomaly detection, or better hardware. If real robot experiments later show that masked training, Bayesian fusion, or independent detectors match the proposed model on paired failures, the claim should be killed.

8 Limitations and terminal decision

The v4.1 evidence is strong enough to justify continued work but not submission. The limitations are material:

- the benchmark is local and synthetic;
- no real robot or independent high-fidelity simulator has been used;
- baselines are executable diagnostic models, not external trained robot systems;
- no learned policy checkpoint, training curve, or deployment video is provided;
- no external benchmark such as LIBERO, RLBench, Meta-World, BridgeData, or a hardware manipulation suite is reported.

The terminal decision for this rebuild is therefore **STRONG_REVISE**. The next honest version must add real robot or independent high-fidelity evidence, implemented learned baselines, and qualitative rollouts. Without that evidence, the paper should not be submitted to ICLR main.

References

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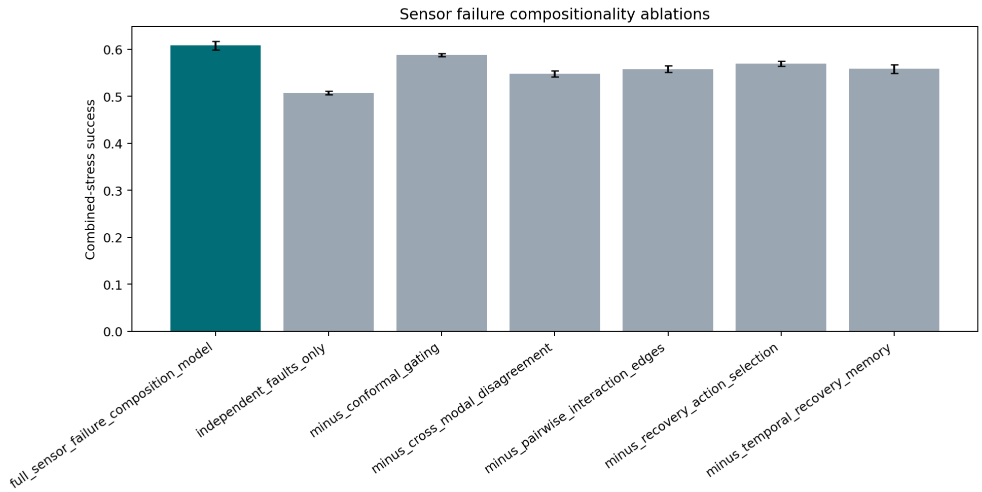

Figure 3: Ablation success under combined stress. The full model remains above all removed-component variants.

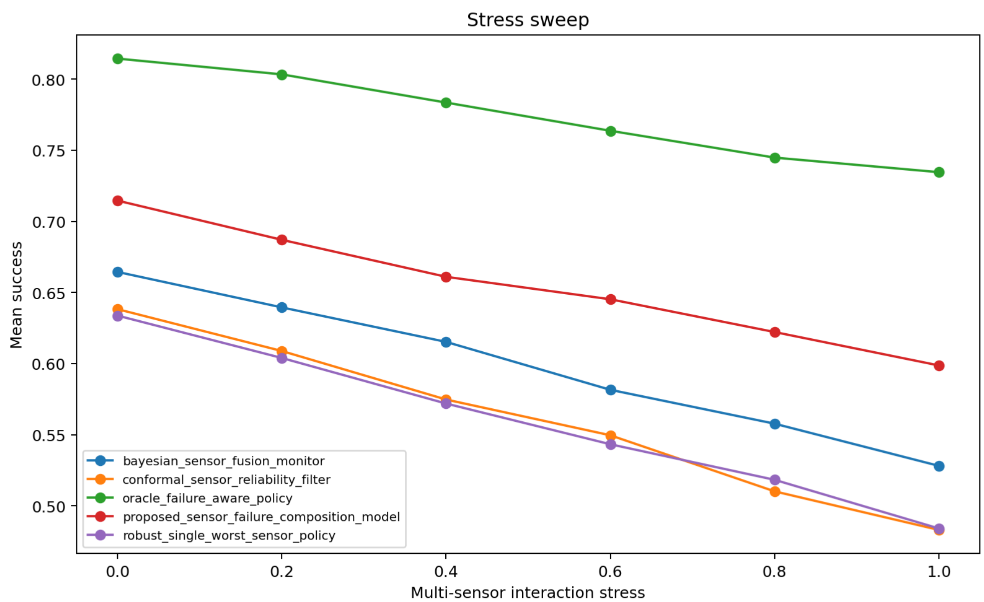

Figure 4: Stress sweep over multi-sensor interaction intensity. The proposed method remains above non-oracle baselines across the generated stress range, while the oracle remains the ceiling.

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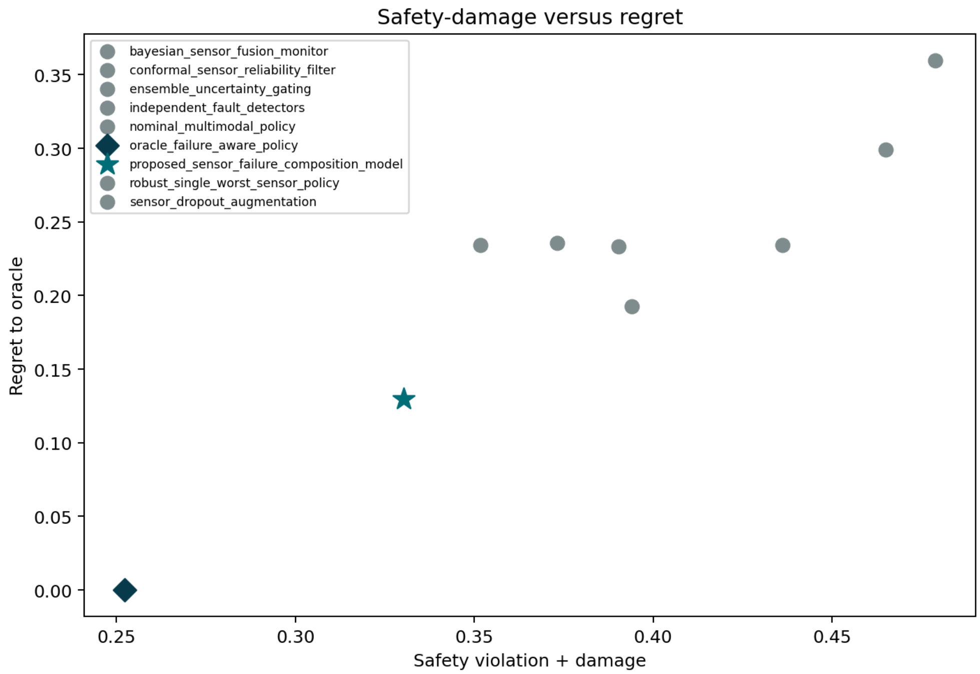

Figure 5: Safety and regret trade-off on combined stress. The proposed method improves success without paying a safety penalty relative to the strongest non-oracle baseline.

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